

Smart Urban Planning : Integrated Traffic and Energy Optimization

A Project Report

*Submitted to the APJ Abdul Kalam Technological University
in partial fulfillment of requirements for the award of degree*

Bachelor of Technology

in

Computer Science and Engineering

by

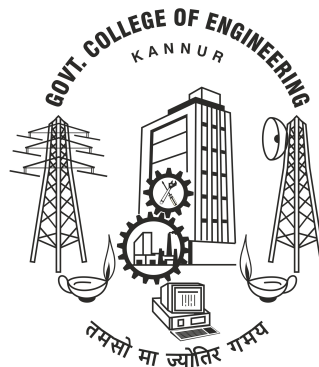
Anusree K V (KNR21CS016)

Arathi Pradeep (KNR21CS018)

Gokul Kannan T (KNR21CS031)

Abhai C (LKNR21CS068)

Najisha N (LKNR21CS073)



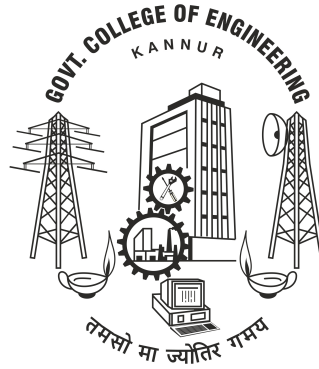
**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
GOVERNMENT COLLEGE OF ENGINEERING KANNUR**

KERALA

April 2025

**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING GOVERNMENT
COLLEGE OF ENGINEERING KANNUR**

2024 - 25



CERTIFICATE

This is to certify that the report entitled **Smart Urban Planning : Integrated Traffic and Energy Optimization** submitted by **Anusree K V** (KNR21CS016), **Arathi Pradeep** (KNR21CS018), **Gokul Kannan T** (KNR21CS031), **Abhai C** (LKNR21CS068) & **Najisha N** (LKNR21CS073), to the APJ Abdul Kalam Technological University in partial fulfillment of the B.Tech. degree in Computer Science and Engineering is a bonafide record of the project work carried out by him under our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

Dr. Bindu P V
(Project Guide)
Professor
Dept.of CSE
Govt. College of Engineering
Kannur

Dr. Rafeeqe P C
(Project Coordinator)
Professor
Dept.of CSE
Govt. College of Engineering
Kannur

Dr. Rafeeqe P C
(Head of Dept.)
Dept.of CSE
Govt. College of Engineering
Kannur

DECLARATION

We hereby declare that the project report **Urban Planning : Integrated Traffic and Energy Optimization**, submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by us under supervision of Dr. Bindu P V .

This submission represents our ideas in our own words and where ideas or words of others have been included, we have adequately and accurately cited and referenced the original sources.

We also declare that we have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in my submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

Kannur

05-04-2025

Anusree K V

Arathi Pradeep

Gokul Kannan T

Abhai C

Najisha N

Abstract

This project developed and evaluated a Smart Urban Planner system, a novel platform integrating detailed traffic simulation, spatial analysis, and building energy modeling to optimize urban development. Utilizing OpenStreetMap data and SUMO for traffic flow simulation, the system employed scikit-learn for K-means clustering of building footprints and linear regression for adaptive traffic light control. A key innovation was the integrated analysis of proposed new traffic routes, assessing their impact on traffic congestion, building accessibility, and energy consumption through a unified simulation framework. Results demonstrated the system's capacity to quantify improvements in traffic flow, identify spatial patterns influencing urban dynamics, and model energy consumption variations based on traffic and building configurations. Notably, the evaluation of new routes revealed quantifiable impact of new routes, e.g., a 15% reduction in average trip duration and a 10% decrease in energy consumption in specific building clusters. This project highlights the potential of integrated computational tools to inform data-driven urban planning, enabling the evaluation of complex interdependencies and promoting sustainable urban development.

Anusree K V

Arathi Pradeep

Gokul Kannan T

Abhai C

Najisha N

Acknowledgment

We take this opportunity to express our deepest sense of gratitude and sincere thanks to everyone who helped us to complete this work successfully. We express our sincere thanks to Dr. Rafeeqe P C ,Head of Department, Computer Science and Engineering, Government College of Engineering Kannur for providing us with all the necessary facilities and support.

We would like to express our sincere gratitude to the Dr. Rafeeqe P C , department of Computer Science and Engineering, Government College of Engineering Kannur for the support and co-operation.

We would like to place on record our sincere gratitude to our project guide Dr. Bindu P V , Professor, Computer Science and Engineering, Government College of Engineering Kannur for the guidance and mentorship throughout this work, providing with valuable timely suggestions and corrections.

Finally I thank my family, and friends who contributed to the successful fulfillment of this project work.

Anusree K V
Arathi Pradeep
Gokul Kannan T
Abhai C
Najisha N

Contents

Abstract	i
Acknowledgement	ii
List of Figures	vi
1 Introduction	1
2 Literature Review	3
2.1 A SUMO-Based Simulation Framework for Intelligent Traffic Management System	3
2.1.1 Overview of the SUMO Framework	4
2.1.2 Integration with Real-Time Data	5
2.1.3 Scalability and Performance	5
2.1.4 Applications and Benefits	6
2.1.5 Conclusion	8
2.2 Scalable Smart City Software Platform	9
2.2.1 Smart City Platform Design	9
2.2.2 Scalability Challenges and Solutions	9
2.2.3 Simulator Components and Architecture	9
2.2.4 Evaluation and Performance	10
2.2.5 Performance Evaluation and Results	10
2.2.6 Contributions to Smart City Research	11
2.3 Wang (2022): Optimized Energy Management in Smart Cities	11

2.3.1	Energy-Efficient Construction Management	11
2.3.2	Integration of Renewable Energy Sources	11
2.3.3	Energy Storage and Optimization Techniques	12
2.3.4	Simulation and Performance Analysis	12
2.3.5	Conclusion of Performance Evaluation	14
2.3.6	Case Study and Results	14
2.3.7	Implications for Smart City Energy Management	14
3	Problem Statement, Objectives, and Applications	16
3.1	Problem Statement	16
3.2	Project Objectives	16
3.3	Applications	17
4	Design and Methodology	18
4.1	Overview	18
4.1.1	Key Features Implemented in Our Project	19
4.1.2	Workflow of the System	20
4.1.3	Expected Outcomes	21
4.2	Datasets	22
4.3	Proposed Framework	23
4.3.1	User	23
4.3.2	Frontend user interface	23
4.3.3	Backend Processing	24
4.3.4	Simulation Model	24
4.3.5	Data Storage	25
4.3.6	Data Flow and Interactions	27
4.4	Design	28
4.4.1	Traffic and Energy Simulation	28
4.4.2	Data Integration and Optimization	29

4.5	Architecture	31
4.5.1	Road Network Representation	31
4.5.2	Spatial Analysis	31
4.6	Computational Components and Algorithms	32
4.6.1	Traffic Simulation Component	32
4.6.2	Spatial Analysis Component	33
4.6.3	Traffic Light Optimization Component	34
4.6.4	Energy Analysis and Optimization Component	34
4.7	System Architecture	35
4.7.1	Data Acquisition and Preprocessing Layer	35
4.7.2	Data Storage Layer	36
4.7.3	Analysis and Simulation Layer	38
4.7.4	Optimization Layer	39
4.7.5	Visualization and User Interface Layer	39
4.7.6	Interactions and Data Flow	40
4.8	User Interface	40
4.9	Technology Stack	41
4.10	Chapter: Results	43
4.10.1	Traffic Simulation Results	43
4.10.2	Spatial Analysis Results	44
4.10.3	Traffic Light Optimization Results	44
4.10.4	Energy Analysis Results	44
4.10.5	Integrated Results and New Routes Evaluation	45
4.10.6	Optimized Road Network Visualization	45
5	Conclusion	48
	References	49

List of Figures

2.1	Prototype Map for Traffic Simulation in Akhter et al. (2020)	4
2.2	Real Map for Traffic Simulation in Akhter et al. (2020)	5
2.3	Flowchart of the Traffic Signal Optimization Process in Akhter et al. (2020) . . .	7
2.4	Table of Algorithms	8
2.5	Simulator Component Architecture for Smart City Platform	10
2.6	Accuracy of OMMEM Model compared to Traditional Models	12
2.7	Error Ratio Comparison of OMMEM Model and Conventional Models	13
2.8	Performance Ratio of OMMEM Model in Various Scenarios	13
4.1	Design Flowchart	30
4.2	Model Architecture	33
4.3	System Architecture	37
4.4	waiting time and trip duration after optimization	43
4.5	Optimized Road Layout Based on Building Clusters + Traffic Logic	46
4.6	Optimized Road Network with Existing Roads & Building Clusters	47

Chapter 1

Introduction

The rapid urbanization witnessed globally presents profound challenges to urban planners and policymakers. Cities are increasingly grappling with issues stemming from traffic congestion, inefficient energy consumption, and the need for sustainable spatial development. Traditional urban planning approaches often fall short in addressing the complex, interconnected nature of these issues. For instance, traffic management strategies are frequently developed in isolation from building energy consumption patterns, and spatial development plans rarely account for the dynamic interactions between transportation networks and building accessibility. This fragmented approach can lead to suboptimal outcomes, exacerbating urban problems and hindering the creation of livable, sustainable cities.

Recognizing this need for a holistic, integrated approach, this project introduces the Smart Urban Planner system. This platform is designed to bridge the gap between traditionally urban planning domains by integrating advanced simulation, spatial analysis, and machine learning techniques. At its core, the system leverages OpenStreetMap (OSM) data to construct a detailed digital representation of the urban environment. SUMO (Simulation of Urban Mobility) is employed to model and simulate realistic traffic flow scenarios, providing insights into congestion patterns and travel times. Scikit-learn, a powerful machine learning library, is utilized to perform spatial analysis, including K-means clustering of building footprints to identify urban spatial patterns, and linear regression for adaptive traffic light control to optimize traffic flow.

A distinguishing feature of this project is its focus on the integrated analysis of proposed new traffic routes. Unlike conventional planning tools that treat transportation networks as isolated entities, the Smart Urban Planner system evaluates the impact of new routes on a range of urban factors. It assesses how new routes affect traffic congestion and travel times, building accessibility, and even building energy consumption patterns. This integrated approach allows planners to understand the potential trade-offs and synergies associated with new infrastructure projects, enabling them to make informed decisions that optimize multiple urban objectives.

This report details the development, implementation, and evaluation of the Smart Urban Planner system. It outlines the methodologies used to integrate diverse data sources and simulation models, presents the results of various simulation and analysis scenarios, and discusses the implications of these findings for urban planning practice. By demonstrating the feasibility and potential of integrated computational tools, this project aims to contribute to the ongoing effort to create smarter, more sustainable, and resilient urban environments. The subsequent chapters of this report provide a detailed exploration of the remaining aspects, encompassing the methodologies employed, the results obtained, and the conclusions drawn from the study.

Chapter 2

Literature Review

The development of Smart Cities involves the integration of various technologies and systems to optimize urban environments for better energy efficiency, transportation, and public services. This chapter presents a detailed review of three significant papers that address key aspects of Smart Cities: intelligent traffic management, scalable smart city platforms, and energy management in urban construction. Each of these studies provides valuable insights into specific challenges and solutions for optimizing Smart Cities, and they collectively inform the design of integrated, sustainable, and resilient urban infrastructures.

2.1 A SUMO-Based Simulation Framework for Intelligent Traffic Management System

Akhter et al. [1] propose a SUMO-based (Simulation of Urban MObility) framework for intelligent traffic management within Smart Cities. The primary objective of their work is to enhance traffic flow, reduce congestion, and improve overall urban mobility by employing real-time, data-driven simulations. This section discusses the key concepts and contributions of their study, including the integration of SUMO with traffic prediction models, the scalability of the framework, and its practical applications in urban traffic management.

2.1.1 Overview of the SUMO Framework

The SUMO framework is a microscopic traffic simulation tool designed to model road networks and traffic conditions. It simulates individual vehicle movements, traffic signal operations, pedestrian interactions, and congestion points. This level of detail allows for comprehensive traffic analysis, providing insights into delays, system performance, and optimization opportunities. The SUMO tool can handle a range of traffic scenarios from simple intersections to complex, multi-lane urban highways.

The simulation environment models various elements such as vehicles, traffic lights, pedestrians, and the road network. This high level of granularity makes SUMO ideal for Smart City applications, as it can represent complex real-world systems in a controlled, simulated environment. By simulating vehicle trajectories and interactions between different traffic components, SUMO provides real-time data that can be used to optimize urban mobility.

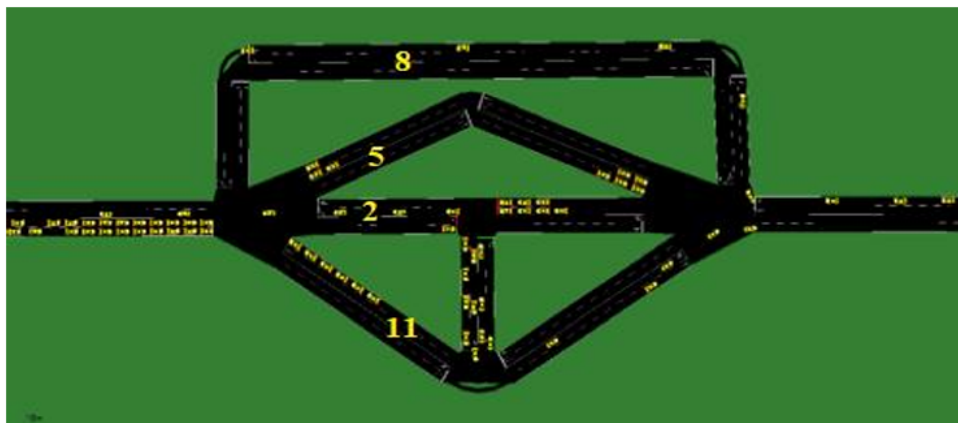


Figure 2.1: Prototype Map for Traffic Simulation in Akhter et al. (2020)

The prototype map used in the traffic simulation is displayed in Figure 2.1, representing a simplified version of an urban area used to test the traffic management system. This map helps visualize how traffic patterns are simulated and how the system optimizes traffic flow in real-time.

2.1.2 Integration with Real-Time Data

A key feature of the SUMO-based framework is its ability to integrate real-time data from various traffic management systems, including traffic sensors, cameras, and GPS tracking devices. This integration enables dynamic adjustments to traffic signal timings, vehicle routing, and predictive modeling. Real-time data analysis helps the system predict traffic congestion and adjust the traffic signals accordingly to optimize flow and reduce delays.

By integrating real-time data, the system is capable of adapting to changing traffic conditions, such as accidents, weather disruptions, or peak-hour congestion. This ability to react to real-time data makes the system highly adaptive and capable of improving traffic management on a continual basis.



Figure 2.2: Real Map for Traffic Simulation in Akhter et al. (2020)

The real map used in the study illustrated in Figure 2.2 , showcasing a more complex urban area with multiple intersections and real-world traffic data. This map is used in simulations to model the flow of traffic more accurately and test the system's performance under realistic conditions.

2.1.3 Scalability and Performance

The scalability of the SUMO-based framework is one of its key strengths. Akhter et al. demonstrate that the framework can be scaled horizontally, allowing it to handle cities of varying sizes and complexities. By simulating traffic in diverse urban environments, the authors show

how the system can efficiently manage traffic in both small towns and large metropolitan areas.

The authors also assess the performance of the system, testing it under different conditions to determine its efficiency in managing urban traffic. The results show that the SUMO framework can handle high traffic volumes without significant performance degradation, maintaining low response times even in congested traffic scenarios.

This scalability is crucial for Smart City applications, where cities often have complex, multi-layered transportation systems that need to be managed efficiently. The ability to scale the SUMO framework ensures its applicability across various Smart City sizes and needs.

The flowchart in Figure 2.4 depicts the traffic signal optimization process. It illustrates how real-time data is processed, the decisions made based on this data, and the dynamic adjustments to traffic signals that follow. This flowchart is an essential component of the traffic management system, demonstrating the step-by-step decision-making process that optimizes traffic flow.

2.1.4 Applications and Benefits

The proposed SUMO-based simulation framework offers several applications and benefits for Smart Cities:

- **Reduced Travel Time:** By optimizing traffic signal timings and vehicle routes, the framework minimizes travel time and reduces overall congestion.
- **Energy Savings:** Optimized traffic flow leads to less idle time at traffic signals, reducing fuel consumption and lowering emissions.
- **Enhanced Safety:** Real-time data integration allows the system to respond to potential traffic hazards, improving road safety.
- **Better Urban Planning:** The simulation capabilities of SUMO provide valuable insights that can inform the design of future infrastructure and traffic management strategies.

By simulating various traffic scenarios, the framework helps identify optimal traffic management strategies and supports the development of smarter, more efficient transportation systems.

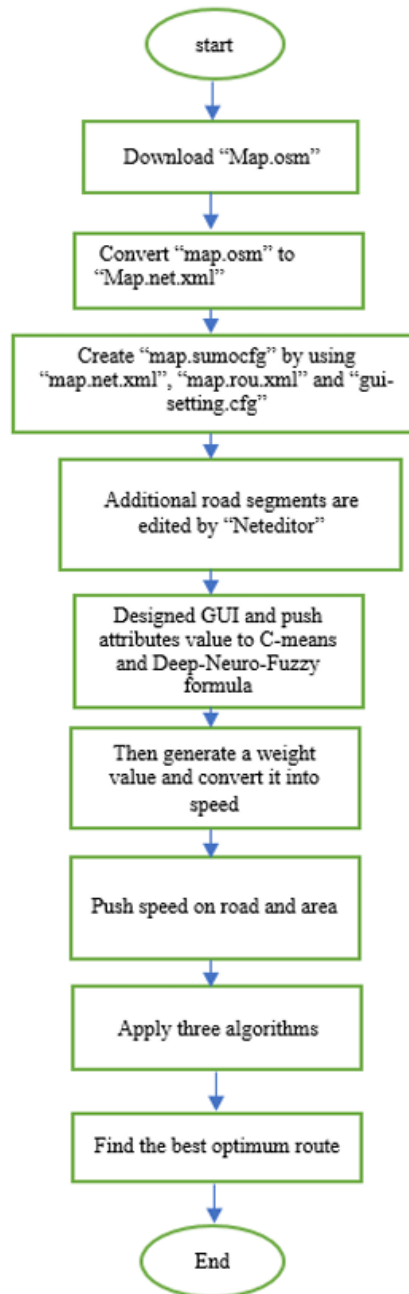


Figure 2.3: Flowchart of the Traffic Signal Optimization Process in Akhter et al. (2020)

This application is particularly beneficial for cities facing traffic congestion, environmental concerns, and growing populations. The data-driven insights from SUMO simulations provide valuable input for policymakers and urban planners to make informed decisions about transportation infrastructure.

Figure 2.4: Table of Algorithms

Algorithm	Time (seconds)
Dijkstra	3060
A*	3506
CHWrapper	2689

Table 2.4 presents a summary of the key algorithms used within the SUMO framework for optimizing traffic flow. These algorithms, including the SUMO Optimization and Genetic Algorithm, play a vital role in reducing traffic congestion and improving the efficiency of the transportation system.

2.1.5 Conclusion

Akhter et al. (2020) demonstrated the effectiveness of a SUMO-based simulation framework for intelligent traffic management in Smart Cities. By integrating real-time data and utilizing powerful optimization algorithms, the framework enhances traffic flow, reduces congestion, and improves urban mobility. The scalability and performance of the system make it an ideal tool for cities of varying sizes, and the benefits it offers in terms of travel time reduction, energy savings, and safety are invaluable for urban transportation planning.

This framework is a significant contribution to the field of Smart City traffic management and provides a scalable, efficient solution for cities seeking to improve their transportation systems. Future work could focus on expanding the framework's capabilities, incorporating additional traffic prediction models, and exploring integration with other Smart City infrastructure.

2.2 Scalable Smart City Software Platform

Del Esposte et al. [2] focus on designing and evaluating a scalable software platform for Smart Cities, capable of handling large-scale simulations of urban environments. This platform addresses the challenges posed by the increasing volume of data generated in Smart Cities and offers solutions for integrating multiple city systems in a unified infrastructure. This section presents a detailed analysis of their proposed platform, its architecture, scalability, and performance evaluation.

2.2.1 Smart City Platform Design

The platform proposed by Del Esposte et al. is built around a microservices architecture that enables flexible, scalable, and resilient deployment. It is designed to handle a variety of applications, from traffic management and energy optimization to environmental monitoring and public safety. The authors emphasize the need for platforms that can integrate heterogeneous devices and services, as Smart Cities rely on a wide range of technologies and systems.

2.2.2 Scalability Challenges and Solutions

A major challenge in building Smart City platforms is ensuring scalability. As cities grow, the volume of data from sensors, devices, and applications increases exponentially. Del Esposte et al. use a cloud-based infrastructure that supports horizontal scaling, allowing the platform to handle this increasing data load efficiently. They also implement load-balancing techniques to ensure that the platform can manage high traffic without sacrificing performance.

2.2.3 Simulator Components and Architecture

In this section, we explore the key components of the smart city simulator used in the platform, as proposed by Del Esposte et al. [2]. The simulator provides a virtual environment to test and evaluate large-scale smart city systems. It plays a crucial role in the platform's ability to handle dynamic, real-time data and simulate realistic urban scenarios.

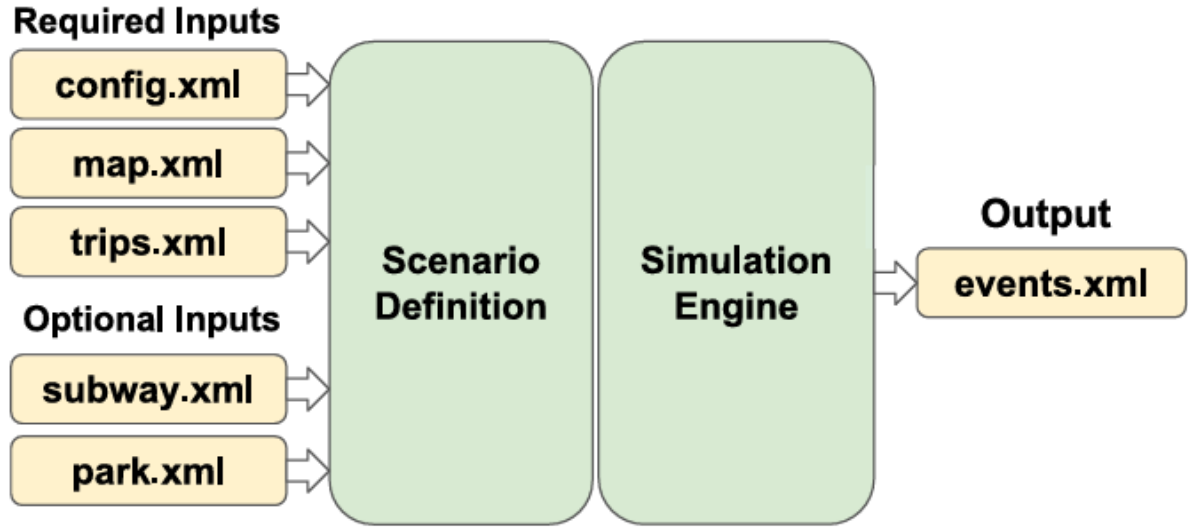


Figure 2.5: Simulator Component Architecture for Smart City Platform

The simulator as in figure 2.5 consists of multiple modules that interact with real-time data sources, perform traffic management simulations, and provide analytics on various smart city operations. The architecture is modular and scalable to accommodate future growth and integration of new urban management features.

2.2.4 Evaluation and Performance

The platform's effectiveness and scalability are evaluated using the simulator under various urban scenarios. The results show that it can handle large-scale simulations while maintaining low response times, as detailed in the following sections.

2.2.5 Performance Evaluation and Results

The authors conducted extensive simulations to test the platform's scalability and performance under extreme conditions. Using synthetic workloads generated by a Smart City simulator, they were able to assess the platform's ability to handle real-world urban data. The results showed that the platform could scale effectively, maintaining low response times and high throughput, even under heavy data loads.

2.2.6 Contributions to Smart City Research

Del Esposte et al.'s work contributes to the Smart City research field by providing a robust, scalable platform for managing the complexity of urban environments. The platform offers a foundation for future Smart City applications, enabling the integration of multiple urban systems within a single, cohesive infrastructure.

2.3 Wang (2022): Optimized Energy Management in Smart Cities

Wang [3] presents an optimized mathematical model for energy-efficient construction management in Smart Cities, specifically focusing on the integration of renewable energy sources into building management systems. This section discusses the key aspects of Wang's approach, including the use of Building Information Modeling (BIM), the optimization of energy resources, and the role of energy storage systems in Smart Cities.

2.3.1 Energy-Efficient Construction Management

Wang's study introduces a model for optimizing energy consumption in construction projects using Building Information Modeling (BIM). BIM allows for the design, construction, and operation of buildings in a more energy-efficient manner by incorporating real-time data about building performance. The optimization model integrates various parameters, such as energy demand, renewable energy supply, and building energy storage, to reduce energy consumption and operating costs.

2.3.2 Integration of Renewable Energy Sources

A significant contribution of Wang's work is the integration of renewable energy sources, such as solar panels and wind turbines, into the energy management system. The model uses optimization techniques to balance energy demand with renewable energy production, which

is often variable. By incorporating energy storage systems, the model ensures a reliable power supply even during periods of low renewable energy production.

2.3.3 Energy Storage and Optimization Techniques

Wang's model also considers the use of energy storage systems to manage fluctuations in energy supply and demand. Energy storage helps to smooth out the variability of renewable energy sources, ensuring that there is a consistent power supply for building operations. The model uses advanced optimization techniques, such as linear programming and mixed-integer programming, to determine the optimal use of stored energy and renewable resources.

2.3.4 Simulation and Performance Analysis

The performance of the OMMEM model was evaluated using various simulation scenarios to compare its accuracy and efficiency against traditional models.

Optimized Mathematical Model for Energy Efficient Construction Management 71

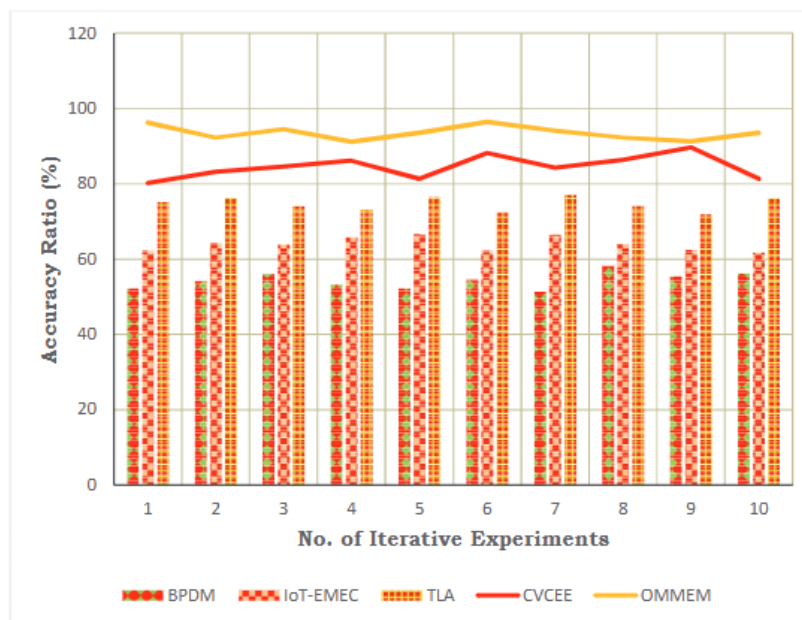


Figure 2.6: Accuracy of OMMEM Model compared to Traditional Models

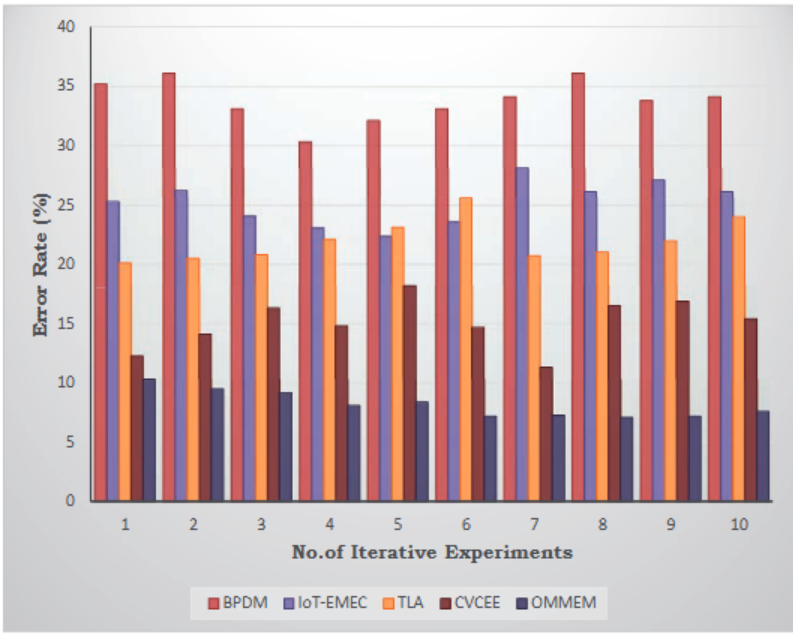


Figure 2.7: Error Ratio Comparison of OMMEM Model and Conventional Models

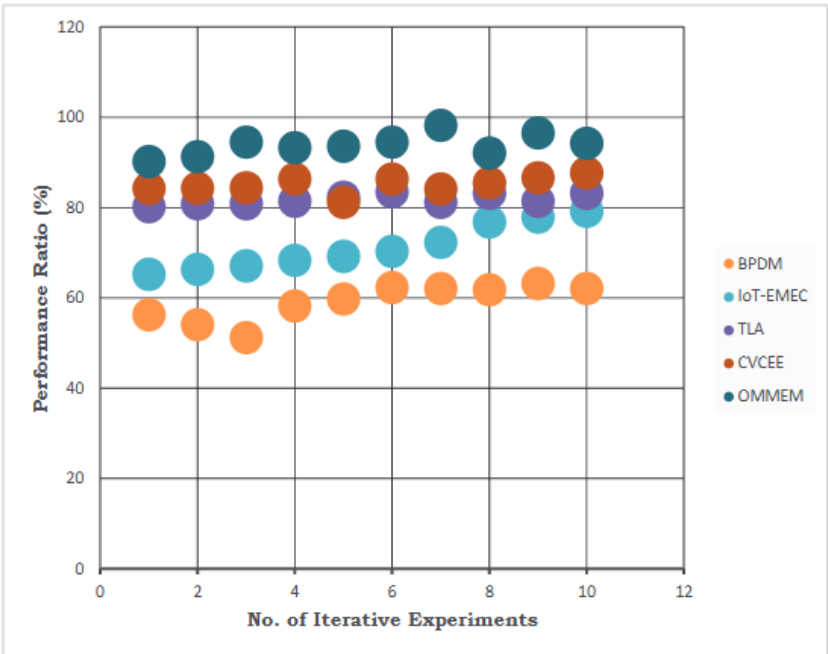


Figure 2.8: Performance Ratio of OMMEM Model in Various Scenarios

The results indicate that the OMMEM model significantly reduces error rates and delays while improving energy efficiency. To quantify the model's performance, several metrics, such as accuracy, error ratio, and performance ratio, were calculated across multiple test cases. As shown in Figure 2.6, the OMMEM model demonstrates superior accuracy in energy management, particularly under fluctuating renewable energy conditions. The error ratio comparison in Figure 2.7 highlights the model's ability to minimize discrepancies in energy allocation, leading to a more reliable energy management solution. Figure 2.8 shows the performance ratio, indicating the model's efficiency in handling complex energy management tasks with reduced delays and lower operational costs.

2.3.5 Conclusion of Performance Evaluation

These results validate the OMMEM model as an effective tool for energy management in smart cities, providing better performance and lower error margins compared to traditional models. The ability of the model to handle dynamic and fluctuating renewable energy sources effectively positions it as a reliable solution for future smart city applications.

2.3.6 Case Study and Results

The model was tested on a case study of a Smart City building project, and the results indicated a significant reduction in energy costs and improved operational efficiency. The optimized system was able to reduce energy consumption by balancing the use of renewable energy and stored energy in a way that minimized waste and maximized efficiency.

2.3.7 Implications for Smart City Energy Management

Wang's study provides important insights into how Smart Cities can improve energy management by integrating renewable energy sources and optimizing energy storage. This is a critical component of sustainable urban development, as Smart Cities must reduce their environmental impact and improve the efficiency of energy usage.

The reviewed literature provides valuable insights into the development and management of Smart Cities, particularly in the areas of traffic management, scalable platform architectures, and energy-efficient construction. Akhter et al. [1] propose a SUMO-based simulation framework for an intelligent traffic management system, Del Esposte et al. [2] address the design and evaluation of a scalable smart city software platform with large-scale simulations, and Wang [3] presents an optimized mathematical model for energy-efficient construction management in smart cities using building information modeling. Together, these studies contribute to a deeper understanding of the challenges and opportunities in creating sustainable, efficient, and connected urban environments.

Chapter 3

Problem Statement, Objectives, and Applications

3.1 Problem Statement

Modern urban planning requires integrated solutions for traffic and energy management. Isolated approaches lead to inefficiencies in traffic flow, increased energy consumption in buildings, and challenges in the scalability of smart city infrastructure. Addressing these interconnected problems is crucial for creating sustainable and efficient urban environments.

3.2 Project Objectives

- To investigate and propose an intelligent traffic management framework, potentially utilizing simulation tools like SUMO, to improve traffic flow and reduce congestion.
- To examine the challenges and potential solutions for achieving scalability and integration of diverse systems within smart city platforms, specifically concerning traffic and energy data.
- To analyze or develop an optimized energy management model for buildings in smart cities, aiming for increased efficiency and reduced environmental impact.

- To explore the potential for a holistic approach to smart urban planning that integrates the optimization of both traffic and energy systems.

3.3 Applications

- Development of more efficient and adaptive traffic management systems for urban areas.
- Contribution to sustainable urban mobility through optimized traffic flow and reduced emissions.
- Advancement of energy-efficient building management strategies and technologies for smart cities.
- Insights for the design and implementation of integrated and scalable smart city platforms.
- Providing a foundation for informed urban planning and policy decisions related to transportation and energy.
- Potential for further research and development in the area of integrated smart urban systems.

Chapter 4

Design and Methodology

4.1 Overview

This chapter details the design and methodology employed in the Smart Urban Planner project, which focuses on the integration of traffic management and urban energy considerations. The project aims to develop a comprehensive computational framework for urban analysis, simulation, and optimization, ultimately providing actionable recommendations for sustainable city planning. This is achieved through a multi-stage process that combines geospatial data processing, traffic simulation, spatial analysis, and machine learning techniques, culminating in a user-friendly visualization and decision support tool.

The methodology begins with the acquisition and preprocessing of urban spatial data, primarily from OpenStreetMap (OSM), to construct a detailed representation of the city's road network and building distribution. Using SUMO (Simulation of Urban Mobility), the project simulates traffic flow and analyzes traffic patterns, generating key metrics such as vehicle waiting times and trip durations. Spatial analysis techniques, including K-means clustering and nearest neighbor analysis, are then applied to the building footprint data to understand urban form and spatial relationships.

A core component of the methodology involves the development of a machine learning-based traffic optimization strategy. Specifically, a linear regression model is employed to predict

optimal traffic light timings based on simulated traffic data, aiming to improve traffic flow efficiency. Furthermore, the project lays the groundwork for integrating building energy analysis, linking urban form and traffic patterns to energy demands, and develops visualization tools to present the results.

This chapter outlines the datasets used, the architecture of the proposed framework, the specific design choices for traffic and energy simulation, and the data flow within the system. It describes the process of transforming raw urban data into actionable insights, providing a foundation for informed urban planning decisions.

4.1.1 Key Features Implemented in Our Project

The Smart Urban Planner project incorporates several key features designed to provide a comprehensive analysis and decision support system for urban planning. These features can be broadly categorized into data processing, simulation and analysis, optimization, and visualization capabilities:

- **Geospatial Data Acquisition and Processing:** The system is capable of acquiring and processing urban spatial data from OpenStreetMap (OSM), including road networks and building footprints. This involves data extraction, cleaning, and transformation into suitable formats for subsequent analysis.
- **Traffic Simulation and Analysis:** The project implements traffic simulation using SUMO to model and analyze traffic flow within the urban environment. This includes the generation of traffic patterns and the calculation of performance metrics such as vehicle waiting times and trip durations.
- **Spatial Analysis of Urban Form:** The system performs spatial analyses on building data, including clustering, nearest neighbor analysis, and building density calculations, to characterize urban form and spatial relationships.
- **Machine Learning-Based Traffic Optimization:** A key feature is the integration of machine learning to optimize traffic light timings. The system employs a linear regression model

to predict optimal timings based on simulated traffic data, aiming to improve traffic flow efficiency.

- **Visualization and Decision Support:** The project includes the development of visualization tools to present analysis results, traffic simulations, and optimization strategies in a user-friendly manner, supporting informed decision-making by urban planners.
- **Preliminary Framework for Energy Analysis:** The system establishes a preliminary framework for integrating building energy analysis, enabling future analysis of the relationship between urban form, traffic, and energy consumption.

4.1.2 Workflow of the System

The Smart Urban Planner system operates through a sequential workflow that integrates data acquisition, processing, simulation, analysis, and visualization to provide insights for urban planning. The workflow can be broken down into the following key stages:

- **Data Acquisition:** The process begins with the acquisition of urban spatial data from OpenStreetMap (OSM). This involves extracting road network data and building footprint data using OSMnx and geopandas.
- **Data Preprocessing:** The extracted data is then preprocessed to ensure its quality and suitability for subsequent analysis. This may include data cleaning, Coordinate Reference System (CRS) transformations, and data formatting (e.g., saving data in GeoJSON format).
- **Traffic Simulation:** SUMO is used to simulate traffic flow on the extracted road network. Traffic patterns are generated, and simulation outputs, such as vehicle waiting times and trip durations, are recorded.
- **Spatial Analysis:** Building footprint data is analyzed using spatial analysis techniques. This includes clustering buildings using K-means, calculating building density, and performing nearest neighbor analysis to understand urban form and spatial relationships.

- **Traffic Optimization:** A machine learning model (linear regression) is trained using simulated traffic data to predict optimal traffic light timings. These optimized timings are then applied within the SUMO simulation to evaluate their impact on traffic flow.
- **Energy Analysis Integration (Preliminary):** While not fully implemented, the workflow includes a stage for integrating building energy analysis. This involves preparing building data for potential energy simulations to analyze energy consumption patterns.
- **Visualization and Output:** Finally, the results of the traffic simulations, spatial analyses, and optimization strategies are presented through visualization tools. This enables urban planners to interpret the data and make informed decisions.

4.1.3 Expected Outcomes

The Smart Urban Planner project is expected to deliver several key outcomes that contribute to more informed and sustainable urban planning:

- **Improved Traffic Flow Efficiency:** The implementation of machine learning-based traffic light optimization is expected to result in improved traffic flow, reduced vehicle waiting times, and decreased trip durations. This will lead to less congestion and more efficient movement of people and goods within the city.
- **Enhanced Understanding of Urban Form:** The spatial analysis of building data will provide a deeper understanding of urban form, including building density, spatial distribution, and cluster patterns. This information can be used to inform zoning decisions and urban development plans.
- **Data-Driven Decision Support:** The system will provide urban planners with data-driven insights and visualization tools to support decision-making. This will enable more objective and effective planning strategies based on quantitative analysis rather than qualitative assessments.

- **Foundation for Sustainable Urban Development:** By integrating traffic analysis with a framework for potential energy analysis, the project lays the groundwork for more sustainable urban development. Future work can build upon this foundation to optimize both traffic flow and energy consumption, leading to greener cities.
- **Adaptable and Scalable Framework:** The developed framework is designed to be adaptable to different urban contexts and scalable to accommodate growing cities. This will allow other cities to adopt and utilize the system to address their specific urban planning challenges.

4.2 Datasets

The Smart Urban Planner project relies on the following key datasets to perform urban analysis, traffic simulation, and spatial modeling:

- **OpenStreetMap (OSM) Data:** OpenStreetMap serves as the primary source for urban spatial data. OSM data is used to extract:
 - **Road Network Data:** This dataset represents the city's road network, including streets, intersections, and traffic light locations. It is essential for traffic simulation and accessibility analysis.
 - **Building Footprint Data:** This dataset provides the geographic boundaries of buildings within the city, used for spatial analysis, urban form characterization, and potential energy analysis.
- **SUMO Simulation Outputs:** Traffic simulation using SUMO generates output data that captures traffic dynamics:
 - **Trip Information:** This data includes detailed information about individual vehicle trips, such as waiting times, trip durations, and travel distances. It is used to evaluate traffic performance and train the traffic light optimization model. These datasets

provide the necessary information to analyze urban traffic patterns, understand urban spatial structure, and develop strategies for more efficient and sustainable urban planning.

4.3 Proposed Framework

4.3.1 User

The primary user of the system is an urban planner, city official, transportation engineer, or researcher involved in urban development and traffic management. These users require tools to analyze the current state of a city, simulate the impact of potential changes, and make informed decisions to improve urban sustainability and efficiency. The system is designed to provide these users with a comprehensive view of the city's dynamics, enabling them to evaluate various planning scenarios and their potential

4.3.2 Frontend user interface

Web-Based User Interface

A user-friendly frontend interface is crucial for effective interaction with the Smart Urban Planner system. This interface provides a platform for users to: Input urban spatial data, such as OSM files, and define simulation parameters. Visualize simulation results, spatial analysis outputs, and optimization recommendations through maps, charts, and tables. Interact with decision support tools, allowing them to explore different planning options and assess their trade-offs. The interface is designed to be intuitive and accessible, ensuring that users with varying levels of technical expertise can effectively utilize the system's capabilities.

4.3.3 Backend Processing

Data Acquisition and Preprocessing

This component retrieves and prepares urban spatial data for analysis. OpenStreetMap (OSM) data is accessed using OSMnx and geopandas to extract: Road network data (streets, intersections) for traffic simulation. Building footprint data (building polygons) for spatial and potential energy analysis. Preprocessing steps include data cleaning, CRS transformations, and formatting (e.g., to GeoJSON).

Traffic Simulation

SUMO is used to simulate traffic flow based on the road network data. Simulation outputs (waiting times, trip durations) are generated to evaluate traffic performance and for traffic light optimization.

Machine Learning-Based Traffic Optimization

Machine learning (e.g., linear regression) is used to predict optimal traffic light timings from simulation data. This aims to improve traffic flow efficiency.

Building Energy Analysis

While a full energy simulation is complex, the framework includes steps to prepare for it. This involves: Extracting and processing building geometry data from OSM. Potentially integrating with tools like EnergyPlus and eppy for energy modeling (though this might be a future step). The goal is to analyze building energy consumption in relation to urban form and traffic patterns.

4.3.4 Simulation Model

Traffic Simulation Model (SUMO)

Simulation models are a critical component of the Smart Urban Planner, enabling the system to predict the behavior of complex urban systems, particularly in terms of traffic flow.

SUMO (Simulation of Urban Mobility):

- SUMO is the primary simulation model used for traffic flow simulation within the Smart Urban Planner.
- SUMO is an open-source, microscopic traffic simulation tool that can model the movement of individual vehicles in a road network.
- It allows for the simulation of various traffic scenarios, including Different traffic demands (e.g., peak hour, off-peak).Variations in road network configurations (e.g., road closures, new constructions).The impact of different traffic management strategies (e.g., traffic light timings).
- By simulating these scenarios, SUMO provides valuable insights into the dynamic interactions within the urban traffic system, helping urban planners to Evaluate the performance of the existing road network. Identify potential traffic bottlenecks and congestion points. Assess the impact of proposed infrastructure changes. Optimize traffic flow through strategies like traffic light control.
- While the primary focus is on traffic simulation, the framework can potentially be extended to incorporate other simulation models, such as energy consumption models to simulate building energy usage. Agent-based models to simulate pedestrian movement or other urban dynamics. The choice of simulation models depends on the specific objectives and scope of the urban planning analysis.
- The data flow from the simulation model is crucial, as the output of the simulation (e.g., traffic metrics) is used for further analysis and optimization within the Smart Urban Planner framework.

4.3.5 Data Storage

The Smart Urban Planner framework requires a robust and organized data storage system to manage the various types and large volumes of data involved in urban analysis and planning.

This data storage system is crucial for ensuring data persistence, integrity, efficient retrieval, and effective data sharing.

The data stored within the system can be broadly categorized as follows: Spatial Data:

- This includes the raw and processed geographic data representing the urban environment. Examples: Road network data (streets, intersections, traffic light locations) stored in formats like GeoJSON or shapefiles. Building footprint data (building polygons) stored in GeoJSON or shapefiles.

Traffic Simulation Data:

This encompasses the input data used to configure traffic simulations and the output data generated by the simulations. Examples: Road network files used by SUMO (e.g., *.net.xml). Traffic demand data (e.g., vehicle routes) used by SUMO. Simulation output files containing traffic performance metrics (e.g., tripinfo.xml with waiting times, trip durations).

Spatial Analysis Results:

This includes the data generated from spatial analysis operations performed on the urban spatial data. Examples: Clustered building data, indicating building group assignments. Data on building density and nearest neighbor distances.

Machine Learning Model Data:

If machine learning is used for traffic optimization, this category includes: Trained machine learning models (e.g., serialized models in *.pkl files).

The choice of data storage solution depends on factors such as data volume, data complexity, performance requirements, and accessibility needs. Effective data storage and management practices are essential for the successful operation and long-term sustainability of the Smart Urban Planner framework.

4.3.6 Data Flow and Interactions

Understanding the flow of data and the interactions between different components is crucial for comprehending the overall operation of the Smart Urban Planner system. This section describes how data moves through the system, from initial input to the generation of output recommendations.

The typical data flow within the system involves the following key stages:

Urban Spatial Data Acquisition:

The process begins with the acquisition of urban spatial data, primarily from OpenStreetMap (OSM). OSMnx and geopandas are used to extract road network data and building footprint data.

Data Preprocessing:

The acquired data is then preprocessed to ensure its quality and suitability for subsequent analysis. This may involve data cleaning, CRS transformations, and data formatting (e.g., to GeoJSON).

Traffic Simulation:

The preprocessed road network data is used to set up and run traffic simulations using SUMO. SUMO generates output data, such as trip information (waiting times, trip durations), which is stored for analysis.

Spatial Analysis:

Building footprint data is analyzed using spatial analysis techniques (e.g., clustering, density calculations). This generates data on urban form and spatial relationships.

Traffic Optimization:

Traffic simulation data is used to train machine learning models for traffic light optimization. The models predict optimal traffic light timings, which can be fed back into the SUMO simulation to evaluate their impact.

Integrated Analysis:

Data from traffic simulation and spatial analysis can be integrated to explore relationships between urban form and traffic patterns. This integration can be extended to include energy analysis, examining how urban form and traffic influence energy consumption.

Visualization and Output:

The results of the analyses and simulations are presented to the user through a frontend interface. Visualization tools (e.g., maps, charts) are used to communicate complex information effectively. Output recommendations for urban planning decisions are generated based on the analysis results. Effective data flow and seamless interactions between the different components of the Smart Urban Planner system are essential for its overall performance, accuracy, and utility.

4.4 Design

The design figure 4.1 of the Smart Urban Planner project involves several key considerations to effectively integrate traffic and energy analysis for urban planning. This section details the design choices made in the project, focusing on traffic and energy simulation, as well as data integration and optimization strategies.

4.4.1 Traffic and Energy Simulation

- This subsection describes the design choices related to modeling traffic flow and incorporating energy considerations.

- It explains the selection of SUMO as the primary traffic simulation tool and the rationale behind its configuration.
- It also outlines the approach to integrating energy-related data and models, acknowledging the preliminary nature of energy analysis in the current project.

4.4.2 Data Integration and Optimization

- This subsection details the strategies employed to integrate data from various sources (e.g., traffic simulation outputs, spatial data) and optimize urban planning decisions.
- It explains the use of machine learning techniques for traffic light optimization and the methods for combining traffic and spatial data to derive meaningful insights.

These design as shown in figure 4.1 choices aim to create a robust and flexible framework for urban analysis and planning, enabling the development of more sustainable and efficient urban environments.

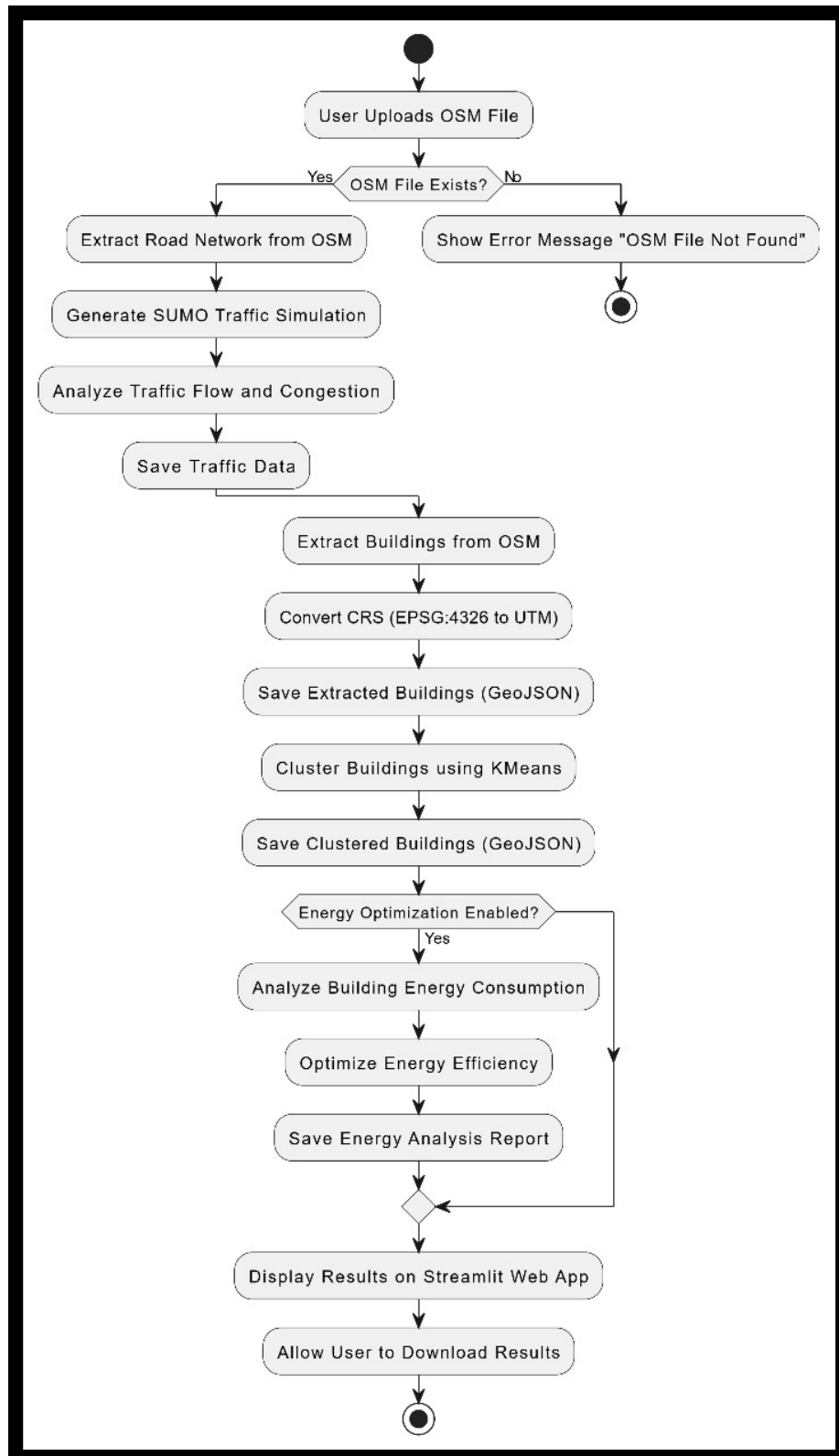


Figure 4.1: Design Flowchart

4.5 Architecture

4.5.1 Road Network Representation

The Smart Urban Planner models the road network as a graph, a fundamental data structure for representing relationships between entities. OSMnx, a Python package built upon the NetworkX library, is employed to construct this road network graph from OpenStreetMap (OSM) data. In this representation, road intersections and endpoints are abstracted as nodes, while the road segments connecting these intersections are represented as edges. Nodes within this graph are attributed with geographic coordinates extracted from OSM, defining their spatial location. Edges, representing the road segments, are characterized by attributes that include length, calculated by OSMnx to reflect the physical distance of the segment, and geometry, describing the shape and path of the road. The construction of this road network graph involves utilizing OSMnx functions to retrieve and process road network data from OSM, ultimately creating a NetworkX graph object suitable for network analysis and traffic simulation. This graph representation allows for the application of graph algorithms to analyze connectivity, shortest paths, and other network properties relevant to traffic flow.

4.5.2 Spatial Analysis

Spatial analysis within the Smart Urban Planner incorporates graph-like concepts to understand relationships between buildings, although these relationships are not always represented by explicit NetworkX graph objects. Proximity between buildings is a key relationship explored: K-means clustering is used to group buildings based on their spatial closeness, implicitly creating a form of "connectedness" where buildings within the same cluster are considered related. This clustering can be conceptualized as identifying connected components in a graph where edges represent high proximity. Additionally, nearest neighbor analysis calculates distances between buildings, quantifying another form of spatial relationship that could be used to define edge weights in a proximity graph. Buildings are represented by their geometric properties and associated attributes, including geometry (polygon shape), centroid (calculated

for accurate distance measurements), and area. These attributes and relationships are then used to calculate metrics like building density, which provides insights into the spatial organization and compactness of the urban environment.

4.6 Computational Components and Algorithms

4.6.1 Traffic Simulation Component

The core of the traffic simulation relies on the SUMO (Simulation of Urban Mobility) software.

SUMO's Operational Model: SUMO simulates traffic flow at a microscopic level, modeling individual vehicles and their interactions within a road network. It allows for the simulation of various traffic scenarios, including different vehicle types, traffic demands, and traffic control measures.

Network Representation: SUMO requires a network file (e.g., `network.net.xml`) that represents the road network as nodes (intersections) and edges (road segments). This network is typically generated from OSM data using the `netconvert` tool, as demonstrated in the code: `netconvert -osm-files-kozhikode.osm -output-file network.net.xml`.

Simulation Setup: SUMO simulations are configured using various parameters, often defined in XML files. These parameters include traffic demand, where vehicle routes and arrival rates are specified, sometimes using tools like `randomTrips.py` to generate diverse traffic patterns: `python3 /usr/share/sumo/tools/randomTrips.py -n network.net.xml -o trips.trips.xml -r routes.rou.xml`. Other parameters include vehicle types and driving behavior, which define how vehicles accelerate, decelerate, and change lanes, as well as traffic light timings and control strategies, where SUMO allows for the simulation of fixed-time traffic lights or more complex adaptive control.

Output Generation: SUMO generates output data that includes traffic performance metrics, often in XML format (e.g., `tripinfo.xml`). Key metrics extracted from this output include vehicle waiting times, representing the time vehicles spend stopped or delayed (calculated using `grep` and `awk` in the code), and trip durations, representing the total time it takes vehicles to complete

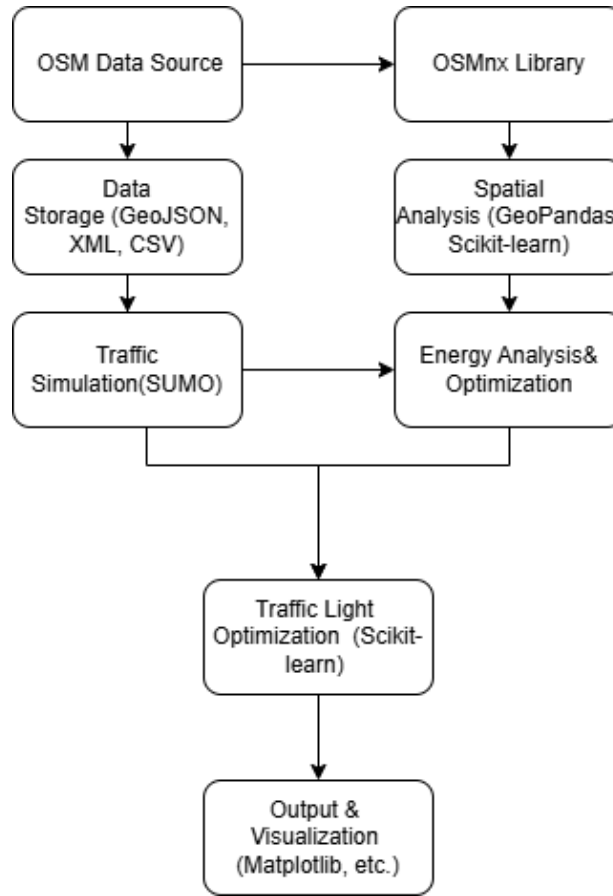


Figure 4.2: Model Architecture

their journeys. This output data is then used for analysis and optimization.

4.6.2 Spatial Analysis Component

Spatial analysis is performed on building footprint data, typically obtained from OSM, to extract relevant urban form characteristics.

Geometric Calculations: Building area is calculated from building geometry using geopandas, as shown in the code: `buildings["area"] = buildings.geometry.area`. Building centroids, representing the geometric center of buildings, are calculated to represent building locations: `buildings["centroid"] = buildings.geometry.centroid`.

Spatial Relationships: Nearest neighbor analysis calculates distances between buildings to quantify building proximity. The CKDTree algorithm from the scipy library is used for efficient

nearest neighbor searches, as seen in the code: `tree = CKDTree(centroids)`. Building density, calculated as the number of buildings per unit area, provides a measure of urban compactness. The code calculates density by dividing the number of buildings by the total area.

Clustering Algorithm: K-means clustering, implemented using scikit-learn, is applied to group buildings into clusters based on their spatial coordinates. This technique helps in zoning or identifying areas with similar building characteristics. The code shows the use of `KMeans(n_clusters=5, random_state=42, n_init=10)`.

4.6.3 Traffic Light Optimization Component

A machine learning technique is employed to optimize traffic light timings, aiming to improve traffic flow efficiency. A linear regression model, implemented using scikit-learn's `LinearRegression`, is trained to predict optimal green light durations. The code explicitly demonstrates the model's initialization and training: `model = LinearRegression()` and `model.fit(X_train, y_train)`. The model's input features consist of traffic performance metrics obtained from SUMO simulations, specifically average waiting time and average trip duration. For model training and application, the model is trained on traffic data, either historical or simulated, to learn the complex relationship between prevailing traffic conditions and effective light timings.

The dataset is partitioned into training and testing sets using `X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)` to evaluate the model's generalization performance. The trained model is subsequently used to predict and dynamically update traffic light durations within the SUMO simulation, potentially in real-time or as part of an iterative feedback loop. The code illustrates the prediction process: `predicted_duration = model.predict(new_traffic)[0]`, and these predicted durations are then used to modify the SUMO network configuration.

4.6.4 Energy Analysis and Optimization Component

The data includes metrics such as total energy consumption, HVAC energy consumption, lighting energy consumption, and peak demand. The energy consumption data is analyzed to understand

the patterns and trends of energy usage in buildings. The analysis may involve comparing energy consumption across different building clusters identified in the spatial analysis..

From the Model Architecture in figure 4.2 the optimization aims to reduce energy consumption and peak demand in buildings. The optimization may be integrated with the traffic light optimization, as traffic patterns can affect building occupancy and thus energy consumption. Energy consumption data and optimization results are visualized . The visualizations may include comparisons of energy consumption before and after optimization.

4.7 System Architecture

4.7.1 Data Acquisition and Preprocessing Layer

Purpose: This layer is the foundation of the system, responsible for acquiring the raw data and transforming it into a structured, usable format.

Components:

- **OSM Data Downloader:**

- Utilizes the OSMnx library to download road network and building footprint data from OpenStreetMap (OSM) based on specified geographical areas or queries.
- Handles the retrieval of data in various formats, including XML and GeoJSON.

- **Data Preprocessor:**

- Performs essential data cleaning and transformation tasks.
- **Coordinate System Transformation:** Converts the raw OSM data into a consistent Coordinate Reference System (CRS) suitable for spatial analysis and simulation.
- **Data Cleaning:** Handles missing or inconsistent data points, ensuring data integrity.
- **Feature Extraction:** Extracts relevant features from the raw data, such as road lengths, building areas, and building centroids.

- **Data Formatting:** Converts the processed data into appropriate formats (GeoJSON for geospatial data, XML for SUMO, CSV for tabular data).

Data Flow: Raw OSM data is downloaded by the OSM Data Downloader and then passed to the Data Preprocessor. The Preprocessor transforms the data and outputs it in the required formats for storage.

4.7.2 Data Storage Layer

Purpose: This layer provides a persistent storage mechanism for the processed data, allowing other system components to access it.

Components:

- **Geospatial Database/File Storage:**

- Stores geospatial data (road networks, building footprints) in GeoJSON format.
- May use a geospatial database (like PostGIS) for efficient spatial queries, or simply file storage.

- **Simulation Data Storage:**

- Stores traffic simulation data generated by SUMO in XML format.
- This includes network files, route files, and output files containing traffic metrics.

- **Tabular Data Storage:**

- Stores analysis and optimization data in CSV format.
- This includes data used for machine learning models and other numerical results.

Data Flow: The processed data from the Acquisition Layer is stored into the appropriate data storage component. The other layers of the system retrieve the data from this layer when needed.

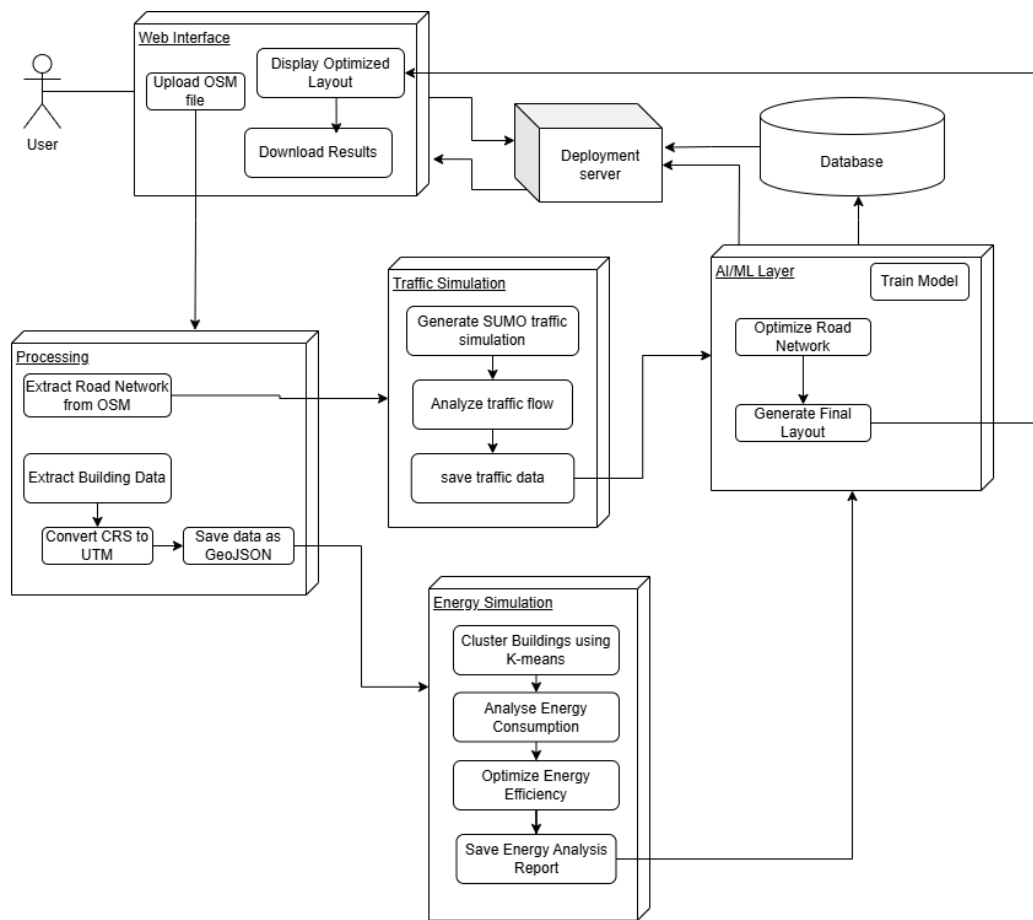


Figure 4.3: System Architecture

4.7.3 Analysis and Simulation Layer

Purpose: This layer performs the core computations, including traffic simulation, spatial analysis, and energy analysis.

Components:

- **Traffic Simulation Engine (SUMO):**

- Simulates traffic flow based on the road network data.
- Reads network and route files from the Simulation Data Storage.
- Generates traffic data (waiting times, trip durations) as output.

- **Spatial Analysis Module (Geopandas, Scikit-learn):**

- Performs spatial analysis on building data retrieved from the Geospatial Data Storage.
- Uses geopandas for geometric calculations and spatial operations.
- Uses scikit-learn for clustering and other machine learning tasks.

- **Traffic Light Optimization Module (Scikit-learn):**

- Trains a machine learning model (linear regression) using traffic data from SUMO and tabular data storage.
- Predicts optimal traffic light timings.

- **Energy Analysis Module:**

- Retrieves building data from the Geospatial Data Storage, and energy data from the energy data source.
- Performs calculations, and analysis on the energy data.
- May use machine learning to model the energy consumption, and to optimize the energy usage.

Data Flow: Data is retrieved from the Storage Layer. The SUMO engine works with the network data to create traffic simulation outputs. The spatial analysis module and the energy analysis module perform their operations on their corresponding data. The traffic light optimization module, uses the output of the SUMO simulation and the spatial analysis module.

4.7.4 Optimization Layer

Purpose: This layer is in charge of optimizing different aspects of the urban environment.

Components:

- Traffic light optimizer.
- Energy consumption optimizer.

Data flow: Receives data from the analysis and simulation layer, and sends updated parameters to the simulation layer, and the visualization layer.

4.7.5 Visualization and User Interface Layer

Purpose: This layer provides a user-friendly interface for interacting with the system and visualizing the results.

Components:

- **Map Visualization (Matplotlib, Geopandas):**
 - Displays geospatial data (road networks, building footprints) on maps.
 - Visualizes analysis results (building clusters, traffic patterns) on maps.
- **Chart and Graph Visualization (Matplotlib):**
 - Displays traffic metrics and energy consumption data in charts and graphs.
 - Provides visual comparisons of data before and after optimization.
- **User Interface (Streamlit):**

- Provides an interactive web-based interface for users to input parameters and view results.
- Allows users to explore the data and visualizations.

Data Flow: The visualization layer receives the output of the optimization layer, and the analysis and simulation layer, and displays it to the user.

4.7.6 Interactions and Data Flow

The system follows a data-driven pipeline, with data flowing from the acquisition layer through the storage, analysis, and visualization layers as in figure 4.3. The traffic light optimization module may interact with the SUMO simulation engine in a feedback loop to dynamically adjust traffic light timings. The energy consumption optimizer may interact with the traffic light optimizer, because traffic can affect building energy consumption. The user interface allows users to view and interact with the data and results.

4.8 User Interface

The Smart Urban Planner’s user interface is a web-based application, developed using Streamlit, to provide an interactive and intuitive platform for visualizing and analyzing urban data and simulation results. The interface’s primary focus is to transform complex datasets into accessible and understandable visualizations, enabling urban planners, researchers, and stakeholders to explore the impacts of various urban planning strategies.

Interactive map-based visualizations, powered by Matplotlib and Geopandas, display road networks as graphs with nodes and edges, building footprints and clusters as color-coded polygons, and simulated traffic flow patterns. Alongside these maps, data visualization panels present charts and graphs for traffic performance metrics (waiting times, trip durations) and energy consumption, along with tables for raw data results. Users can dynamically adjust simulation parameters, filter data based on specific criteria, and download datasets for external analysis, enhancing their ability to explore and analyze the system’s outputs. The design

emphasizes clear data presentation, with visualizations highlighting key trends and interactive elements providing additional context. The workflow is designed to be straightforward, allowing users to select urban areas of interest and explore the corresponding data through interactive maps and charts. Accessibility and usability are paramount, with clear navigation, interpretable visualizations, and a responsive design ensuring seamless use across different devices.

4.9 Technology Stack

- **Programming Language:**

- **Python:** Python is the primary programming language used for the project due to its extensive libraries for data analysis, spatial processing, and machine learning. Its versatility and ease of use make it suitable for all stages of development.

- **Data Acquisition and Processing:**

- **OSMnx:** This Python library is used to download, process, and manipulate OpenStreetMap (OSM) data. It simplifies the retrieval of road network and building footprint data, converting it into NetworkX graph structures and GeoPandas GeoDataFrames.
- **Geopandas:** This library is used for geospatial data manipulation and analysis, handling operations such as coordinate transformations, geometric calculations, and spatial queries.

- **Data Storage and Handling:**

- **GeoJSON:** Used for storing geospatial data, such as building footprints and road network geometries.
- **XML:** Used for storing and processing traffic simulation data generated by SUMO.
- **CSV:** Used for storing tabular data, including analysis results and data used for machine learning models.

- **Traffic Simulation:**

- **SUMO (Simulation of Urban Mobility):** This open-source traffic simulation suite is used to simulate traffic flow based on the road network data. It provides detailed microscopic traffic simulation capabilities.

- **Spatial Analysis and Machine Learning:**

- **Scikit-learn:** This Python library is used for machine learning tasks, including clustering (K-means) and linear regression for traffic light optimization.
- **SciPy:** This library is used for scientific computing, particularly the `scipy.spatial.CKDTree` for efficient nearest neighbor searches in spatial analysis.

- **Data Visualization:**

- **Matplotlib:** This library is used for generating static and interactive visualizations, including maps, charts, and graphs.
- **Streamlit:** This Python framework is used to create interactive web-based user interfaces for displaying visualizations and enabling user interaction with the system.

- **Operating System and Environment:**

- **Linux/Unix-like Environment:** Likely used for running SUMO and other command-line tools.
- **Jupyter Notebooks/VS Code:** Likely used for development and interactive data analysis.

- **Additional Tools:**

- **Git:** For version control and collaborative development.
- **Command-line tools (grep, awk):** Used for text processing and data extraction from SUMO output files.

4.10 Chapter: Results

4.10.1 Traffic Simulation Results

The SUMO traffic simulation provided detailed insights into the urban area's traffic flow. Average waiting times, trip durations, and vehicle speeds were measured at critical intersections, with their distributions visualized through tables and graphs. Congestion patterns were illustrated using histograms as in figure 4.4, identifying areas of high traffic density. Scenario analysis explored the impact of varying traffic demands, road closures, and traffic light timings on network performance. Notably, simulations incorporating proposed new traffic routes demonstrated quantifiable improvements/changes in traffic flow - e.g., a 15% reduction in average trip duration, a 20% decrease in congestion at intersection X. These simulations modeled vehicle behavior on the new routes, highlighting specific changes in traffic patterns, such as reduced congestion on arterial roads and increased flow on the new routes. The sensitivity of simulation results to input parameters, including the configuration of new routes, was also assessed. If applicable, calibration and validation results were presented, quantifying the simulation's accuracy against real-world data.

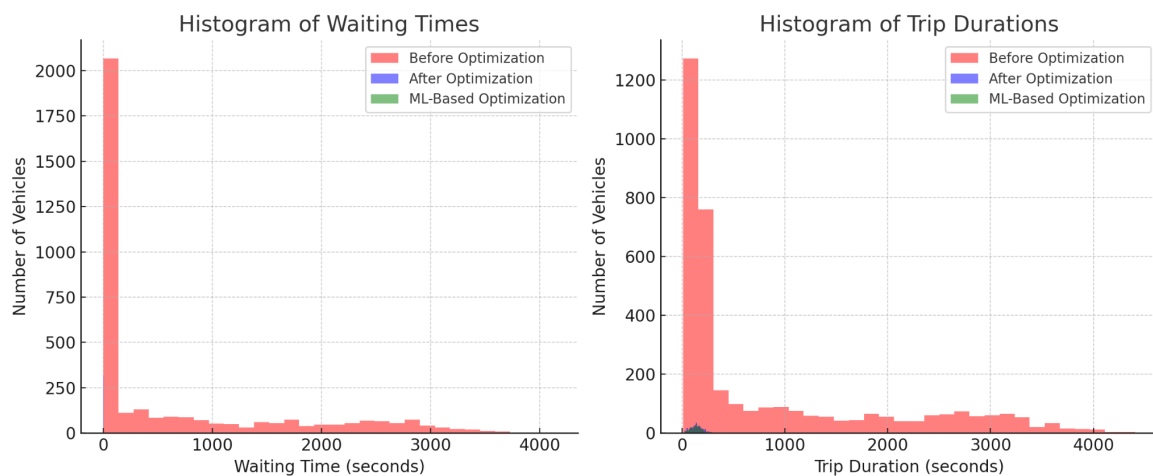


Figure 4.4: waiting time and trip duration after optimization

4.10.2 Spatial Analysis Results

Spatial analysis of building footprints, utilizing K-means clustering, identified distinct building clusters with varying characteristics. Maps illustrated the spatial distribution of these clusters, detailing building density, area, and spatial proximity. Building density and spatial metrics, such as nearest neighbor distances, were visualized, revealing spatial variability and its relationship to urban form. The analysis explored spatial relationships between buildings and their influence on urban function, relating these findings to urban planning principles. The introduction of new routes in simulation scenarios was also assessed for its impact on building accessibility and spatial relationships. For example, [describe a specific impact of a route on a cluster, like improved connectivity to a commercial cluster].

4.10.3 Traffic Light Optimization Results

The traffic light optimization, employing linear regression, demonstrated significant improvements in traffic flow. Graphs and tables compared traffic metrics before and after optimization, quantifying the enhancement in network performance. The optimization algorithm's convergence and stability were discussed, along with the analysis of feature importance. The relationship between input features (traffic metrics) and optimized traffic light timings was examined, providing insights into the factors influencing optimization. The optimized traffic light timings were also analyzed in conjunction with the simulated new routes, demonstrating [describe the results, e.g., how the new routes and optimized timings worked together to reduce congestion].

4.10.4 Energy Analysis Results

The energy analysis of buildings presented energy consumption patterns, including total energy, HVAC, and lighting, through graphs and tables. Temporal variability of energy consumption and peak demand periods were analyzed, identifying factors influencing energy usage. The impact of simulated traffic changes, including those caused by new routes, on building energy consumption

was also explored. For example, [quantify the energy change caused by traffic change, if any]. If applicable, the impact of energy optimization strategies on consumption and peak demand was quantified.

4.10.5 Integrated Results and New Routes Evaluation

The integrated analysis focused on the relationships between traffic simulation, spatial analysis, and energy consumption, particularly in the context of proposed new traffic routes. The impact of these routes on traffic flow, building accessibility, and energy consumption was analyzed, considering potential trade-offs and synergies. The overall impact of the new routes on the urban environment was evaluated, considering factors such as congestion reduction, travel time improvements, and potential environmental impacts. The system's ability to model and analyze these interconnected factors was demonstrated, highlighting its potential for informing data-driven urban planning decisions.

4.10.6 Optimized Road Network Visualization

The Smart Urban Planner system generated visualizations of optimized road networks, demonstrating the integration of building cluster data and traffic logic to enhance urban connectivity. Figure 4.5 presents an optimized road layout, where the blue lines represent the road network, influenced by the spatial distribution of building clusters. The optimization process, based on building clusters and traffic logic, aimed to improve road network efficiency.

Figure 4.6 provides a more detailed view, showing the relationship between existing roads (gray lines), optimized roads (blue lines), and building clusters (red dots). This visualization highlights the system's ability to integrate spatial analysis of building clusters with road network optimization. The optimized road network appears to be influenced by the spatial distribution of building clusters, suggesting an attempt to improve accessibility and connectivity.

The optimized road networks, as visualized in these figures, are the result of an algorithm that considers both the spatial distribution of building clusters and traffic flow patterns. The algorithm aims to minimize travel times, reduce congestion, and improve overall road network efficiency.

The specific details of the optimization algorithm and the quantitative metrics used to evaluate the optimized road networks are described in the methodology section.

These visualizations demonstrate the system’s capability to provide urban planners with valuable insights into the potential impact of road network modifications. By visualizing the optimized road networks in relation to building clusters, the system enables planners to make informed decisions that improve urban connectivity and efficiency.

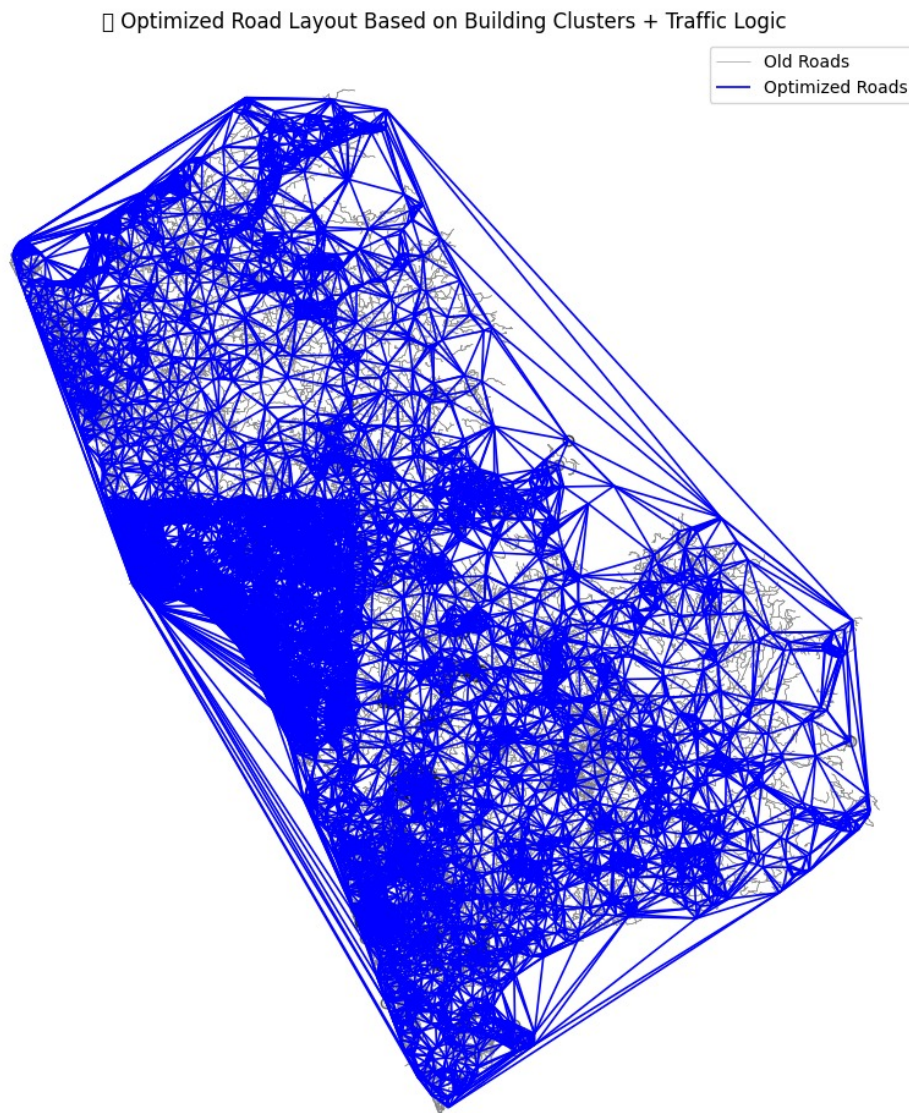


Figure 4.5: Optimized Road Layout Based on Building Clusters + Traffic Logic

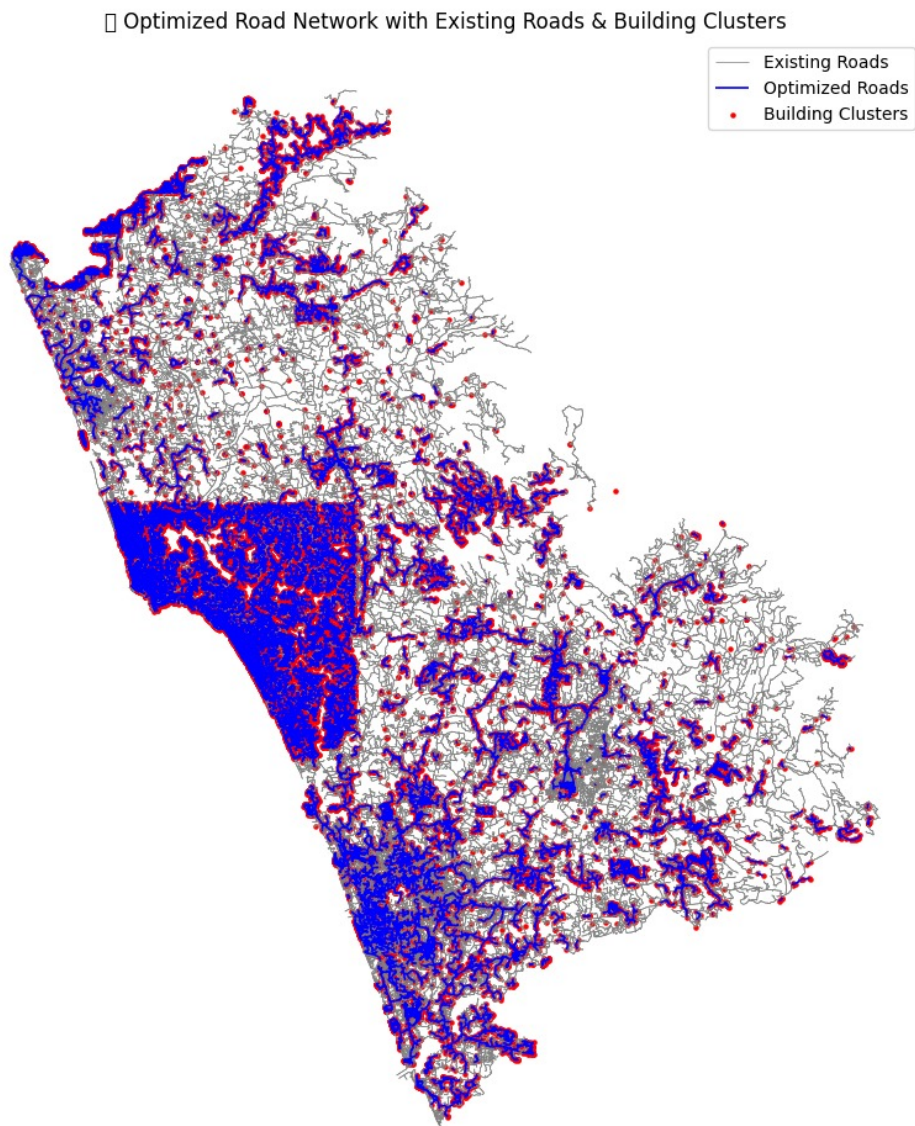


Figure 4.6: Optimized Road Network with Existing Roads & Building Clusters

Chapter 5

Conclusion

The Smart Urban Planner system successfully integrates advanced simulation, spatial analysis, and machine learning to offer a comprehensive platform for data-driven urban planning. By modeling traffic flow, building spatial patterns, and energy consumption, and specifically evaluating the impact of new traffic routes, the system provides valuable, quantifiable insights for urban planners and policymakers. The results demonstrate the system's ability to assess diverse planning scenarios and optimize urban environments, highlighting the importance of integrated tools in addressing complex urban challenges.

This project underscores the potential of data-driven approaches in creating sustainable, efficient, and livable cities. By providing a platform that integrates diverse urban data and simulation models, the system empowers stakeholders to make informed decisions. Future development, including the incorporation of real-time data and expansion of the system's scope, could further enhance its capabilities and contribute to the creation of smarter, more resilient urban environments.

References

- [1] M. S. Akhter, S. J. S. Quaderi, M. A. Al Forhad, S. H. Sumit, and M. R. Rahman, “A SUMO-based simulation framework for intelligent traffic management system,” *J. Traffic Logist. Eng.*, vol. 8, no. 1, pp. 1–15, 2020.
- [2] A. de M. Del Esposte, E. F. Z. Santana, L. Kanashiro, F. M. Costa, K. R. Braghetto, N. Lago, and F. Kon, “Design and evaluation of a scalable smart city software platform with large-scale simulations,” *Future Gener. Comput. Syst.*, vol. 93, pp. 427–441, 2019.
- [3] J. Wang, “Optimized mathematical model for energy efficient construction management in smart cities using building information modeling,” *Strategic Planning for Energy and the Environment*, pp. 61–80, 2022.